

ANALYTICS PRIMER

DATA ANALYST SKILLS & CONCEPTS

A practical guide to the modern data analytics toolkit

SQL · EXCEL · PYTHON · POWER BI · STATISTICS

What We'll Cover



The Analyst's Role

How raw data becomes a business decision



Core Toolkit

Five pillars analysts rely on daily



SQL

Querying & structuring data



Excel & Reporting

Spreadsheets to MIS dashboards



Python

Pandas, NumPy & visualization



Power BI

Modeling, DAX & dashboards



Statistics & EDA

Turning numbers into insight



Career Roadmap

Fresher to expert, step by step

What Does a Data Analyst Do?

Analysts repeat one core cycle daily — turning raw, messy data into clear, actionable decisions for the business.



Collect

Pull from databases, files & APIs



Clean

Fix errors, gaps & duplicates



Analyze

Find patterns, test hypotheses



Visualize

Build charts & dashboards



Communicate

Present insight, recommend action

60–80%

of an analyst's time typically goes into preparing and cleaning data — long before the real analysis begins.

The Analyst's Toolkit

Five core pillars every data analyst builds on



SQL

Extract & query relational data



Excel

Quick analysis & MIS reporting



Python

Automate analysis at scale



Power BI / Tableau

Model & visualize for stakeholders



Statistics

Validate patterns with rigor



Communication

Translate insight into action

SQL — Querying & Structuring Data



SELECT, WHERE, GROUP BY, HAVING

Filtering and aggregating raw tables



JOINS — Inner, Left, Right, Full

Combining data across related tables



Window Functions

RANK, ROW_NUMBER, LAG/LEAD without collapsing rows



CTEs & Subqueries

Breaking complex logic into readable steps



Query Performance

Indexing basics for queries that scale

```
-- Top 3 products by revenue, per region
WITH ranked_sales AS (
  SELECT region, revenue,
         RANK() OVER (
           PARTITION BY region
           ORDER BY revenue DESC
         ) AS rnk
  FROM sales
)
SELECT * FROM ranked_sales
WHERE rnk <= 3;
```

Applied in: Amazon SQL Business Analysis — 20+ queries

Excel & MIS Reporting



Pivot Tables & Pivot Charts

Summarizing large data in seconds



VLOOKUP / XLOOKUP / INDEX-MATCH

Joining data across sheets & workbooks



Conditional Formatting & Validation

Cleaner, error-resistant spreadsheets



Macros & Basic VBA

Automating repetitive MIS tasks



Dashboard-style MIS Reports

Daily & weekly stakeholder reporting



Sample weekly MIS report

Applied in: E-Commerce Sales Dashboard

Python for Data Analysis

```
import pandas as pd

df = pd.read_csv('bookings.csv')

summary = (
    df.groupby('city')['price']
      .agg(['mean', 'count'])
      .sort_values('mean', ascending=False)
)
print(summary.head(10))
```

Applied in: Airbnb Bookings EDA — 50,000+ rows



Pandas

DataFrames — groupby, merge, pivot_table



NumPy

Array operations & numerical computing



Matplotlib & Seaborn

Static visualizations for analysis



Jupyter Notebook

Iterative, documented exploration

Power BI & Data Visualization



Power Query

Cleaning & shaping data before modeling



Data Modeling

Star schema & relationships between tables



DAX Measures

Calculated metrics like YoY growth, attrition rate



Interactivity

Slicers, drill-through & bookmarks



Storytelling

One dashboard, one clear message



Sales Trend



Attrition Mix

18.7%

Attrition Rate



YoY Growth

Applied in: HR Attrition Dashboard — Power BI + DAX

Statistics for Analysts



Descriptive Statistics

Mean, median, mode, standard deviation



Probability & Distributions

Normal curve, spread, likelihood



Correlation vs Causation

Related does not mean caused by



Hypothesis Testing

p-values, confidence intervals, A/B tests



Normal distribution — the backbone of statistical inference

Exploratory Data Analysis (EDA)



Understand the Data

Shapes, columns, data types & source context



Clean & Prepare

Handle missing values, duplicates & outliers



Univariate Analysis

Look at the distribution of each variable alone



Bivariate Analysis

Explore relationships between pairs of variables



Summarize Insights

Surface patterns, anomalies & the next question

Data Cleaning Best Practices

Common Issue	Best-Practice Fix
Missing values	Impute (mean/median) or flag & exclude deliberately
Duplicate records	De-duplicate using a unique business key
Inconsistent formats	Standardize dates, text casing & units
Outliers	Investigate the cause before capping or removing
Wrong data types	Explicitly cast columns (string, integer, datetime)

Translating Data into Business Metrics

The same skillset, different KPIs depending on the domain



Sales & E-Commerce



Revenue



Average Order Value



Conversion Rate



HR & Workforce



Attrition Rate



Headcount



Average Tenure



Operations



Cycle Time



Utilization Rate



Cost per Unit

Beyond the Tools: Communicating Insights



Data Storytelling

Lead with the insight, not the chart



Know Your Audience

Executives want conclusions; analysts want methods



Visual Clarity

Right chart type, minimal clutter



Documentation

Clear READMEs & case studies for reproducibility

“A good analyst doesn’t just find the answer — they make sure the right people understand it.”

Skill Roadmap: Fresher to Expert

1 Beginner

- ✓ Excel fundamentals
- ✓ SQL basics (SELECT, JOIN)
- ✓ Simple charts & summaries
- ✓ Descriptive statistics



2 Intermediate

- ✓ Power BI / Tableau
- ✓ Python with Pandas
- ✓ Window functions & CTEs
- ✓ A/B testing basics



3 Advanced

- ✓ Statistical modeling
- ✓ Workflow automation
- ✓ Advanced DAX & data modeling
- ✓ Stakeholder storytelling

Keep Learning, Keep Analyzing

Turning Data Into Decisions



github.com/sagar-shende



linkedin.com/in/sagarshende-ai